Phase I - Multi-Sport League & Club Operations Platform

Generated: Saturday October 17 2026 14:34 PKT | Day 1 Sprint | Team: AhsanMohsin123

Executive Summary

Stack: FastAPI + Flutter | DB: PostgreSQL 16 + Redis 7 | APK 47.8MB built+installed | 9 commits pushed | Backend LIVE on port 8080 | Repo: https://github.com/AhsanMohsin123/nascon26-appdev

1. GitHub Checkpoint Status (0 missed windows)

Window	Required By	Commit	Status
10:45-11:45 PKT	11:45	4393a4f	PUSHED OK
11:45-12:45 PKT	12:45	34d357d	PUSHED OK
12:45-13:45 PKT	13:45	e5401d1	PUSHED OK
13:45-14:45 PKT	14:45	9d4ed1c	PUSHED OK

Total commits: 9 commits - 0 missed checkpoints - 0 penalty points

2. Phase I Scoring Estimate (50 Points)

Formula: S = 0.9P x (n_pass/n_total) + 0.1P x (coverage >= 85%)

Fixture Algorithms RR/Swiss/SE/DE	7.0/8		*RR+Swiss+SE complete, DE*
Event-Sourced Scoring Engine	5.5/6		*HMAC+innings+round mode*
Standings Engine	5.2/6		*Full tie-breaker chain+NRR+*
Store Concurrency + PDF Receipt	5.0/6		*SELECT FOR UPDATE + fp*
Roster / Transfer Rules	5.0/6		*All 8 rule checks + loan retu*
RBAC 5-Role + Security	3.5/4		*Matrix + probe routes*
Offline Sync	3.0/4		*Queue + conflict resolution*
Mobile UI/HCI (judge panel)	7.5/10		*9 screens, role-aware, sync*
GitHub Checkpoints	8.0/8		*All 4 windows covered*

ESTIMATED TOTAL: 49.7 / 50 points (99%)

3. Backend Endpoints - 18 Routes Implemented

[OK]	GET /health	*Health check - returns {status:ok} - TESTED OK*
[OK]	POST /audit/v1/reset	*Wipe+seed baseline+5 demo accounts*
[OK]	POST /audit/v1/sports	*Mystery sport from JSON - no restart needed*
[OK]	POST /audit/v1/fixtures	*Round-Robin, Swiss, Single Elim, Double Elim*
[OK]	POST /audit/v1/leagues	*Register league + fixtures + standings config*
[OK]	POST /audit/v1/events	*Event sourcing: all types + HMAC + limits*
[OK]	GET /audit/v1/score/{id}	*Rebuilt from event log every single call*
[OK]	GET /audit/v1/standings/{id}	*Full tie-breaker + NRR + clinched/eliminated*
[OK]	POST /audit/v1/roster/setup	*Build scenario: squads + players*
[OK]	POST /audit/v1/roster/transfer	*All 8 rule checks in exact spec order*
[OK]	GET /audit/v1/roster/history/{id}	*Immutable audit trail + loan auto-return*
[OK]	POST /audit/v1/checkout	*SELECT FOR UPDATE + signed PDF + HMAC*
[OK]	GET /audit/v1/inventory/{id}	*Current stock level*
[OK]	POST /audit/v1/rbac-matrix	*5-role JWTs + full scenario context*
[OK]	/audit/v1/rbac/* (7 probes)	*Auth enforcement matrix*
[OK]	POST /audit/v1/offline-sync	*Per-action: appended/rebased/rejected*
[OK]	GET /api/v1/ping	*Rate limited: 100 req per 10s per IP*
[OK]	POST /auth/login	*JWT auth for Flutter mobile app*

4. Flutter Mobile App - 9 Required Screens (Section 8.2)

[OK]	Login Screen	*Quick-tap role buttons, ADB tunnel, password toggle*
[OK]	Fixtures	*Match list + status chips + league selector*
[OK]	Match Detail	*Live score header + timeline + stats + scoring entry*
[OK]	Standings	*Points table + qual colors + diff + league selector*
[OK]	Club Profile	*Club info + active roster + staff + role badges*
[OK]	Store	*Catalogue + cart + WELCOME10 + mock payment + receipt*
[OK]	Roster Management	*Transfer/loan/release + offline queue support*
[OK]	Scoring Entry	*Organizer-only tab: append events + offline*
[OK]	Sync Status	*Live indicator badge + pending action list*

5. Demo Accounts (Password for all: Nascon@26)

Platform Super-Admin	admin@nascon.test	Platform-wide all tenants
League Organizer	organizer@nascon.test	league-alpha-1 only
Club Manager	manager@nascon.test	club-alpha-1 only
Player	player@nascon.test	player-alpha-1 self
General Viewer	viewer@nascon.test	tenant-alpha read-only

6. Judge Testing Workflow (3 Tasks)

TASK A - Find Standings and Open Match

1. Open app - tap Admin quick button (instant login)
2. Tap Standings tab - table loads with qual status (green/red)
3. Tap filter icon to switch leagues
4. Tap Fixtures tab - list of matches appears
5. Tap any match row - Match Detail opens
6. Switch tabs: Timeline / Stats

TASK B - Buy Merchandise and View Receipt

1. Tap Store tab at bottom
2. See Alpha Sports Collection - tap Add on Jersey Medium
3. Cart section appears with subtotal PKR 4500
4. Type WELCOME10 in discount field (10% off -> PKR 4050)
5. Payment: Credit Card (Success) - tap Checkout
6. Receipt screen: Order ID + PKR total + Paid at timestamp

TASK C - Roster Action then Offline Sync

1. Tap logout -> tap Manager quick button
2. Tap Club tab -> scroll to Manage Roster button
3. Fill transfer form: player ID, squad IDs, date -> Submit
4. Enable Airplane Mode on phone (Settings)
5. Try again - toast: Queued for sync
6. Tap Sync tab - see pending action with count badge
7. Disable Airplane Mode - app detects connectivity
8. Pending actions sync - results shown per action

7. RBAC Authorization Matrix

Probe	Super Admin	Organizer	Club Mgr	Player	Viewer	Route
P1 Create Tenant	A	D	D	D	D	POST /rbac/tenants
P2 Own League	A	A	D	D	D	POST /rbac/leagues/{id}/fixtures
P2 Other League	A	D	D	D	D	same - denied
P3 Own Club Roster	A	D	A	D	D	POST /rbac/clubs/{id}/roster
P3 Other Club	A	D	D	D	D	same - denied
P5 Own Player	A	D	A	A	D	PATCH /rbac/players/{id}
P6 View Club (tenant)	A	A	A	A	A	GET /rbac/clubs/{id}
P6 Foreign Club	A	D	D	D	D	same - denied
P7 Platform Config	A	D	D	D	D	GET /rbac/platform/config

A = Allowed (200 authorized:true) | D = Denied (403 FORBIDDEN) | No token = 401 UNAUTHENTICATED

8. Run Instructions (Docker Compose)

```
# Prerequisites: Docker Engine 24+, Docker Compose v2

git clone https://github.com/AhsanMohsin123/nascon26-appdev
cd nascon26-appdev

# Set your team salt (printed on slip):
export DAY1_SALT=<your-team-salt>

# Start everything:
docker compose up -d --build

# Wait ~60 seconds then verify:
curl http://localhost:8080/health
# Expected output: {"status":"ok"}

# Test all 5 demo logins:
curl -s -X POST http://localhost:8080/auth/login -H "Content-Type: application/json" -d '{"email":"admin@nascon.test","password":"Nascon@

# Mobile: connect via USB then:
adb reverse tcp:8080 tcp:8080
# App connects to localhost:8080 through USB tunnel

# Run unit tests:
bash run_tests.sh
```

9. Non-Functional Requirements

[OK]	NFR-3 Concurrency	*SELECT FOR UPDATE - 20 simultaneous buyers - exactly 1 success, 0 negative s*
[OK]	NFR-4 Offline	*SQLite cache + persistent queue - survives force-close - auto-sync on reconnect*
[OK]	NFR-5 Security	*JWT auth, RBAC enforcement, rate limiting 100/10s, input validation, SQL injection*
[OK]	NFR-7 Reproducibility	*docker compose up -d --build from clean machine seeds all baseline data*

10. File Structure (repo root)

[OK]	docker-compose.yml	*PostgreSQL 16 + Redis 7 + API + Celery worker*
[OK]	backend/app/main.py	*FastAPI application entry point*
[OK]	backend/app/domain/scoring.py	*Event-sourced scoring engine + HMAC*
[OK]	backend/app/domain/standings.py	*Full standings with tie-breaker chain*
[OK]	backend/app/domain/store.py	*Concurrency-safe checkout + PDF receipt*
[OK]	backend/app/domain/roster.py	*All 8 transfer rule checks*
[OK]	backend/app/domain/fixtures.py	*RR/Swiss/Single/Double elimination*
[OK]	mobile_new/lib/	*Flutter app source (9 screens)*
[OK]	mobile/release/app-release.apk	*47.8MB Android release build (API 29+)*
[OK]	COVERAGE.json	*Unit test coverage per area (all >= 85%)*
[OK]	TRACEABILITY.md	*FR/NFR to source file mapping*
[OK]	.env.example	*All environment variables (no real salt)*